Year 7
Mathematics

Unit 1: Analysing and Displaying Data

1.2 More Averages and Range

More Averages and Range

STARTER · 5 MIN

Date: _____ **Title: More Averages and Range**

LEARNING OBJECTIVES

- To organise and interpret data using tally charts and frequency tables

STARTER *Answer in your book. No calculators.*

1. Which of 12, 20, 9, 19 are in the class 10–19?
2. Find the mode: 3, 5, 3, 8, 3
3. Find the range: 7, 2, 11, 5
4. Work out 8 + 7 + 3 + 2

Starter — Answers

STARTER

Tick or correct in green pen

1. **12 and 19**
2. **3**
3. **11 – 2 = 9**
4. **20**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use tally marks to record data accurately	☐
SILVER	I can complete a frequency table from a set of data	☐
GOLD	I can interpret a frequency table to answer questions	☐

Tick each box in the plenary when you can do it on your own.

Introduction — More Averages and Range

INTRODUCTION · 3 MIN

A tally chart records data as you count. A frequency table shows how many.

Tally marks

Draw lines in groups of five: the 5th line crosses the other four.

 **= 13**

Frequency

How many times a value (or class) appears.

Total frequency = number of values

Grouped data

Put many different values into equal classes.

1–5, 6–10, 11–15, 16–20

Modal class

The class with the highest frequency.

Discrete: counted. Continuous: measured.

Examples 1 and 2 — Tally Charts and Frequency Tables

EXAMPLES · 5 MIN

BRONZE *Watch and copy into your book*

Hint Tally in groups of five Frequency = how many

Example 1

Put these values into a grouped frequency table:

7, 3, 8, 11, 2, 21, 15, 12, 4, 20, 13, 2, 15, 12, 4, 17

Use the classes 1–5, 6–10, 11–15, 16–20 and 21–25.

Example 2

The table shows pulse rates.

How many had a pulse rate from 90 to 109?

Pulse rate	Frequency
70–79	1
80–89	8
90–99	7
100–109	3

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Tally each value, then count

Class	Tally	Frequency
1–5	𝍸	5
6–10	||	2
11–15	𝍸 |	6
16–20	||	2
21–25	|	1

Tip: check the frequencies add up to 16.

Example 2

Add the two classes

90–99 and 100–109

Students = 7 + 3

= 10

Tip: 90 to 109 covers two classes.

Task 1

PRACTICE · 9 MIN

BRONZE *Use the data to answer each question*

Hint Tally in groups of five Frequency = how many

1. Shoe sizes: 4, 5, 5, 6, 4, 5, 7
   How many students wear size 5?

2. Values: 2, 7, 4, 9, 6, 3, 8, 1
   How many are in the class 1–5?

3. Values: 12, 18, 7, 14, 11, 19
   How many are in the class 10–14?

4. Pulse rates: 70–79: 1, 80–89: 8, 90–99: 7. How many students?

5. Using question 4, how many had a pulse rate of 80 or more?

★ Ages: 11, 12, 11, 13, 12, 11, 12, 11
Which age has the highest frequency?

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 **3**

2 **2, 4, 3, 1**
4 values

3 **12, 14, 11**
3 values

4 **Total = 1 + 8 + 7**
= 16 students

5 **Total = 8 + 7**
= 15 students

★ **11 (frequency 4)**

Examples 3 and 4 — Discrete, Continuous and Modal Class

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Counted: discrete Measured: continuous

Example 3

Is each set of data discrete or continuous?

(a) the numbers of songs on some phones

(b) the masses of some loaves of bread

Example 4

The table shows heights.

Write down the modal class.

Height, h (cm)	Frequency
$120 \leq h < 130$	1
$130 \leq h < 140$	4
$140 \leq h < 150$	6
$150 \leq h < 160$	5
$160 \leq h < 170$	2

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Counted or measured?

(a) Songs are counted:
discrete

(b) Masses are measured:
continuous

Tip: time, length and mass are continuous.

Example 4

Find the highest frequency

The highest frequency is 6.

Modal class = $140 \leq h < 150$

Tip: give the class, not the frequency.

Task 2

PRACTICE · 9 MIN

SILVER

Discrete or continuous? Then use the table.

Hint Counted: discrete Measured: continuous

Wingspans of some birds

Wingspan, w (cm)	Frequency
$0 \le w < 10$	2
$10 \le w < 20$	4
$20 \le w < 30$	6
$30 \le w < 40$	3

1. Is shoe size discrete or continuous?
2. Is the time to run 100 m discrete or continuous?
3. Is wingspan discrete or continuous?
4. Write down the modal class.
5. How many birds were measured?

★ Can you give the largest wingspan?

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

Wingspans of some birds

Wingspan, w (cm)	Frequency
$0 \le w < 10$	2
$10 \le w < 20$	4
$20 \le w < 30$	6
$30 \le w < 40$	3

1. **Discrete: it is counted in sizes**
2. **Continuous: time is measured**
3. **Continuous: length is measured**
4. **$20 \le w < 30$**
5. **Total = 2 + 4 + 6 + 3**
   = 15 birds

★ **No: only its class, $30 \le w < 40$**

Examples 5 and 6 — Grouped Frequency Diagrams

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Modal class: highest frequency
Bars touch for continuous data

Example 5

(2 marks)

Draw a frequency diagram for the heights.

Height, h (cm)	Frequency
$120 \le h < 130$	1
$130 \le h < 140$	4
$140 \le h < 150$	6
$150 \le h < 160$	5
$160 \le h < 170$	2

Example 6

(1 mark)

Use the same table.

How many are at least 150 cm?

Height, h (cm)	Frequency
$120 \le h < 130$	1
$130 \le h < 140$	4
$140 \le h < 150$	6
$150 \le h < 160$	5
$160 \le h < 170$	2

Examples 5 and 6 — Corrections

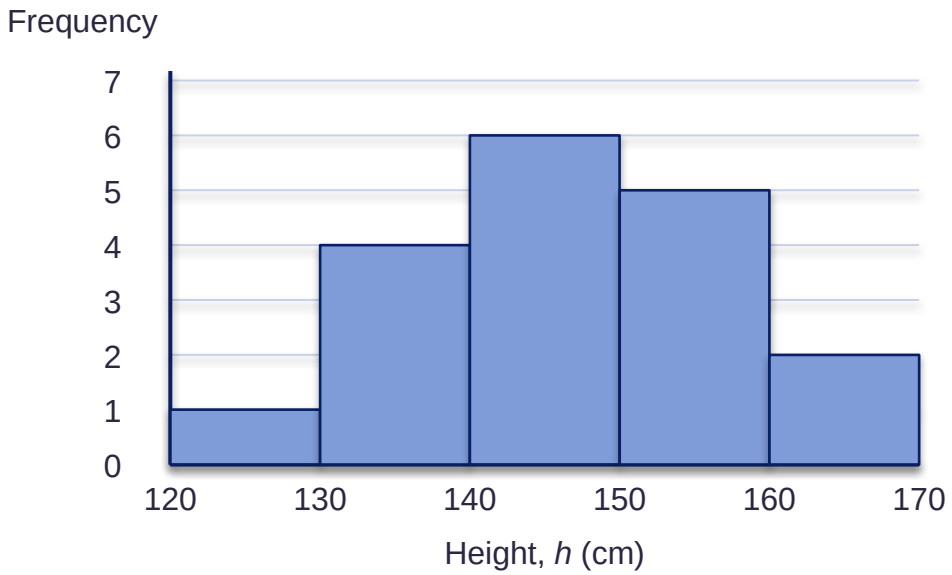
CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Bars touch: height is continuous



Tip: label both axes.

Example 6

'At least 150' means 150 or more

$150 \leq h < 160$ and $160 \leq h < 170$

Students = 5 + 2

= 7

Tip: 150 cm is in the class $150 \leq h < 160$.

Task 3

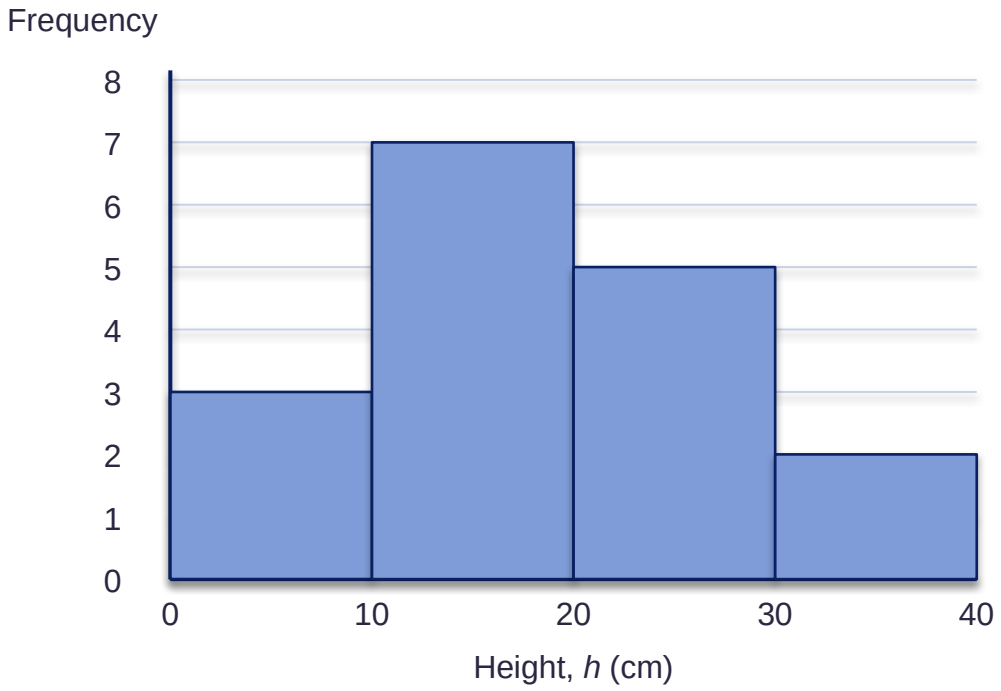
PRACTICE · 9 MIN

GOLD

Use the frequency diagram of plant heights

Hint Modal class: highest frequency
Bars touch for continuous data

Heights of some plants



1. How many plants are $0 \leq h < 10$?
2. Write down the modal class.
3. How many plants were measured?
4. How many plants are at least 20 cm?
5. How many plants are less than 20 cm?

★ What fraction are at least 30 cm?

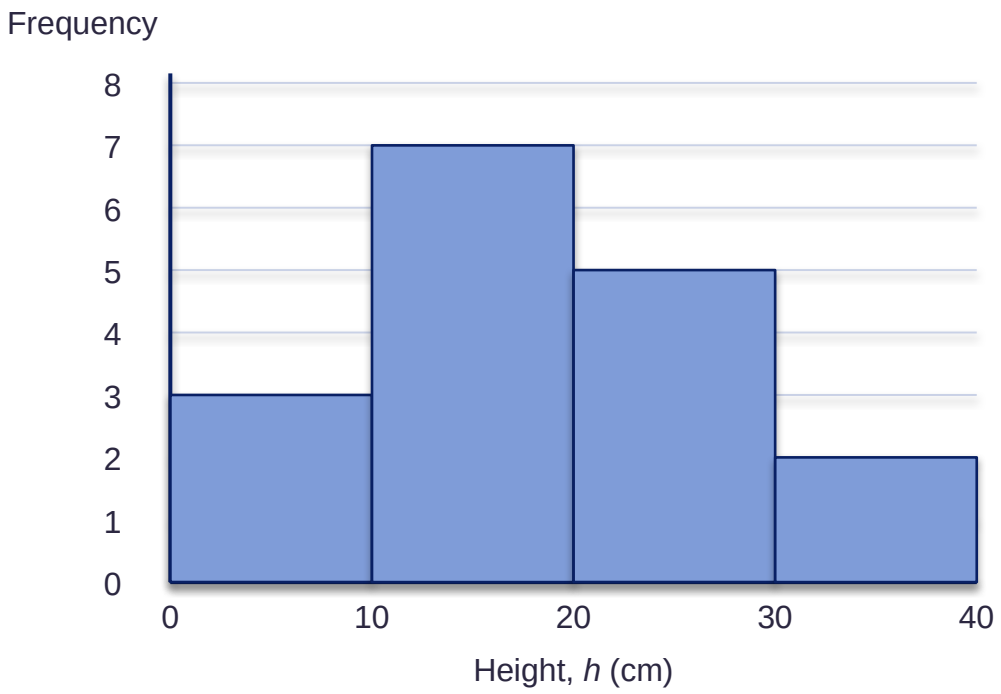
Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

Heights of some plants



1. **3**

2. **$10 \leq h < 20$**

3. **Total = 3 + 7 + 5 + 2**
   = 17

4. **Plants = 5 + 2**
   = 7

5. **Plants = 3 + 7**
   = 10

★ **2 out of 17**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Values 3, 1, 3, 2, 3, 1: what is the frequency of 3?

SILVER Is the time taken to swim 50 m discrete or continuous?

GOLD Classes 0–9: 4, 10–19: 7, 20–29: 2. Write down the modal class.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **3**

SILVER **Continuous: time is measured**

GOLD **10–19**

Exam Practice — Frequency Tables

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Tally in groups of five
Frequency = how many

Question 1

(3 marks)

Goals scored in 12 matches:

2 0 1 3 2 1 0 2 4 2 1 2

Goals	0	1	2	3	4
Frequency					

(a) Complete the frequency table. (2)

(b) Write down the mode. (1)

.....

Question 2

(2 marks)

The table shows the number of pets some students have.

Pets	0	1	2	3
Frequency	5	8	4	3

(a) How many students are there? (1)

.....

(b) How many have at least 2 pets? (1)

.....

Exam Practice — Grouped Data

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Counted: discrete Measured: continuous

Question 3

(3 marks)

Masses, in grams, of 10 potatoes:

412 258 335 225 419

337 315 320 444 180

Mass, M (g)	Tally	Frequency
$100 < M \leq 200$		
$200 < M \leq 300$		
$300 < M \leq 400$		
$400 < M \leq 500$		

(a) Complete the table. (2)

(b) Write down the modal class. (1)

.....

Question 4

(2 marks)

Is each set of data discrete or continuous?

(a) the number of pupils in a class (1)

.....

(b) the time taken to run 100 m (1)

.....

Exam Practice — Frequency Diagrams

EXAM PRACTICE · 10 MIN

GOLD

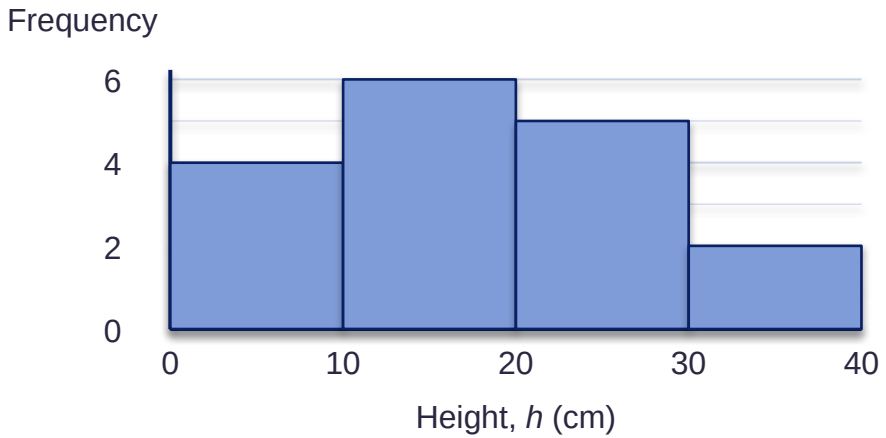
IGCSE exam-style: show all your working

Hint Modal class: highest frequency
Bars touch for continuous data

Question 5

(2 marks)

The diagram shows plant heights.



(a) Write down the modal class. (1)

.....

(b) How many are at least 20 cm? (1)

.....

Question 6

(3 marks)

The table shows the masses of some potatoes.

Mass, M (g)	Frequency
$100 < M \leq 200$	3
$200 < M \leq 300$	5
$300 < M \leq 400$	8
$400 < M \leq 500$	2

(a) How many potatoes are there? (1)

.....

(b) How many are over 300 g? (2)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 0: 2, 1: 3, 2: 5, 3: 1, 4: 1 B2

B1 for 3 correct

(b) Mode = 2 goals B1

Tip: the frequencies add up to 12.

Question 2

(a) Total = 5 + 8 + 4 + 3

= 20 B1

(b) At least 2 = 4 + 3

= 7 B1

Tip: 'at least 2' means 2 or more.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 1, 2, 4, 3 — B2

B1 for tallies or 2 correct

(b) $300 < M \leq 400$ — B1

Tip: 1 + 2 + 4 + 3 = 10 potatoes.

Question 4

(a) discrete — B1

(b) continuous — B1

Tip: counted = discrete, measured = continuous.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) $10 \leq h < 20$ — B1

(b) Plants = 5 + 2

= 7 — B1

Tip: 'at least 20' includes 20.

Question 6

(a) Total = 3 + 5 + 8 + 2

= 18 — B1

(b) Over 300 g = 8 + 2 — M1

= 10 — A1

Tip: use the classes above 300 g.